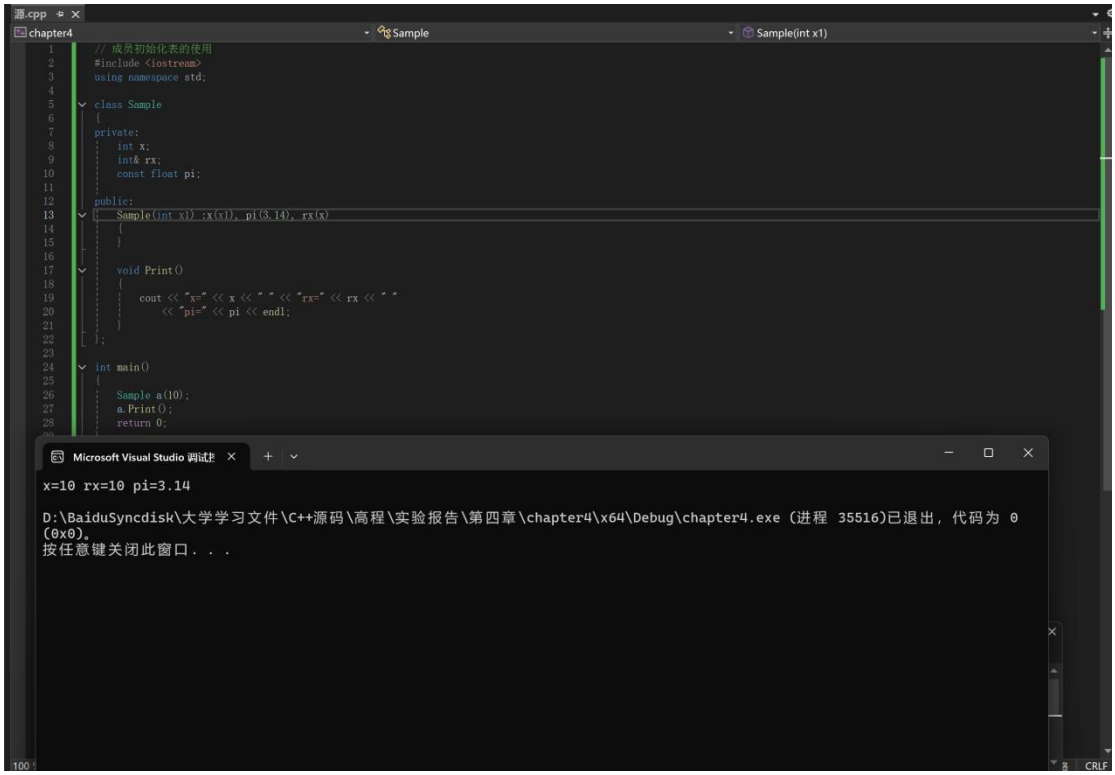
第四章

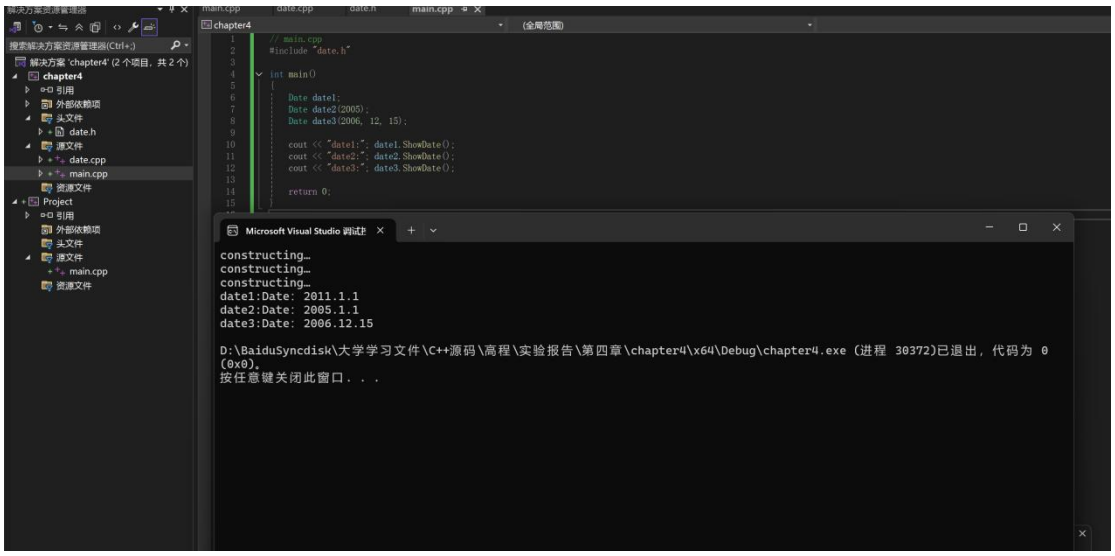
练习 1



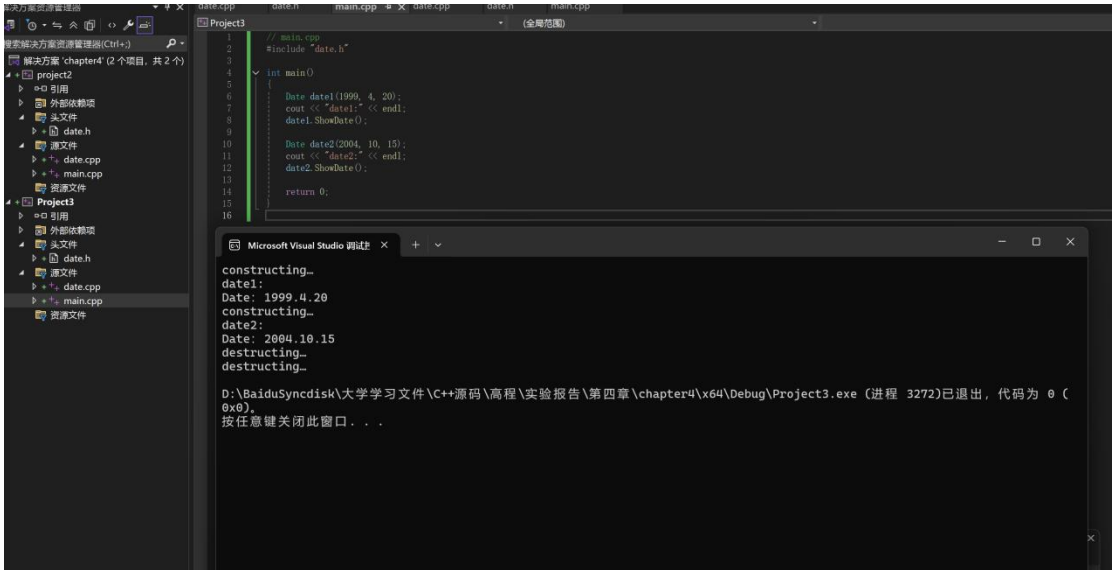
初始化顺序：x->rx->px

必须初始化，普通变量可延迟赋值，引用类型必须在定义时绑定对象，不能先定义后赋值，const 常量只能在定义时初始化，不能在函数体内赋值

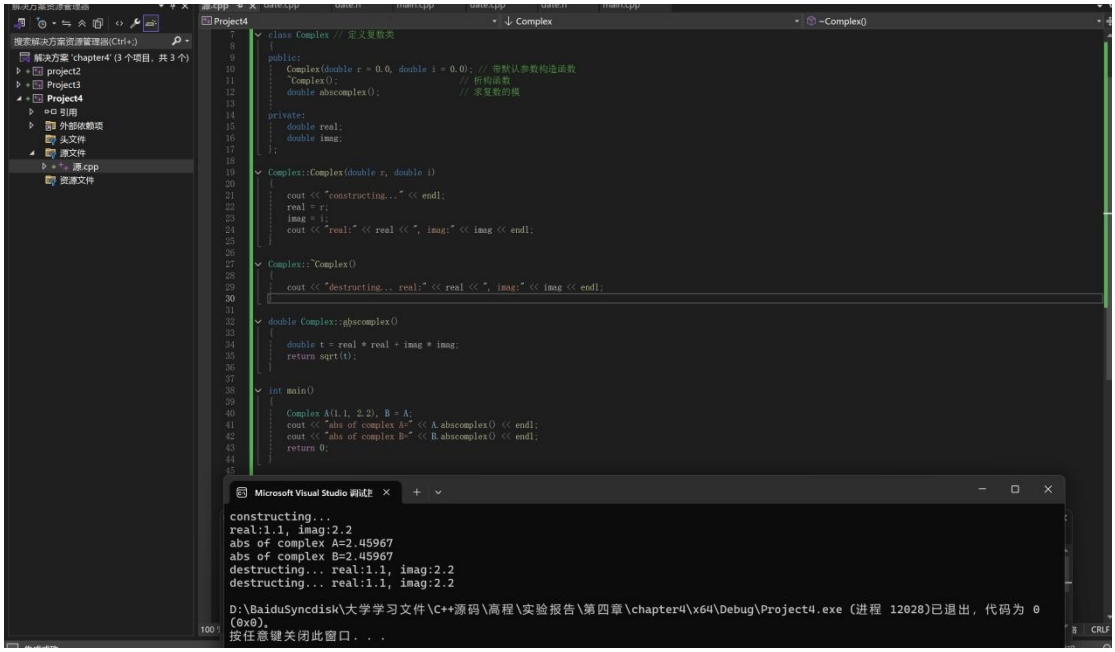
练习 2



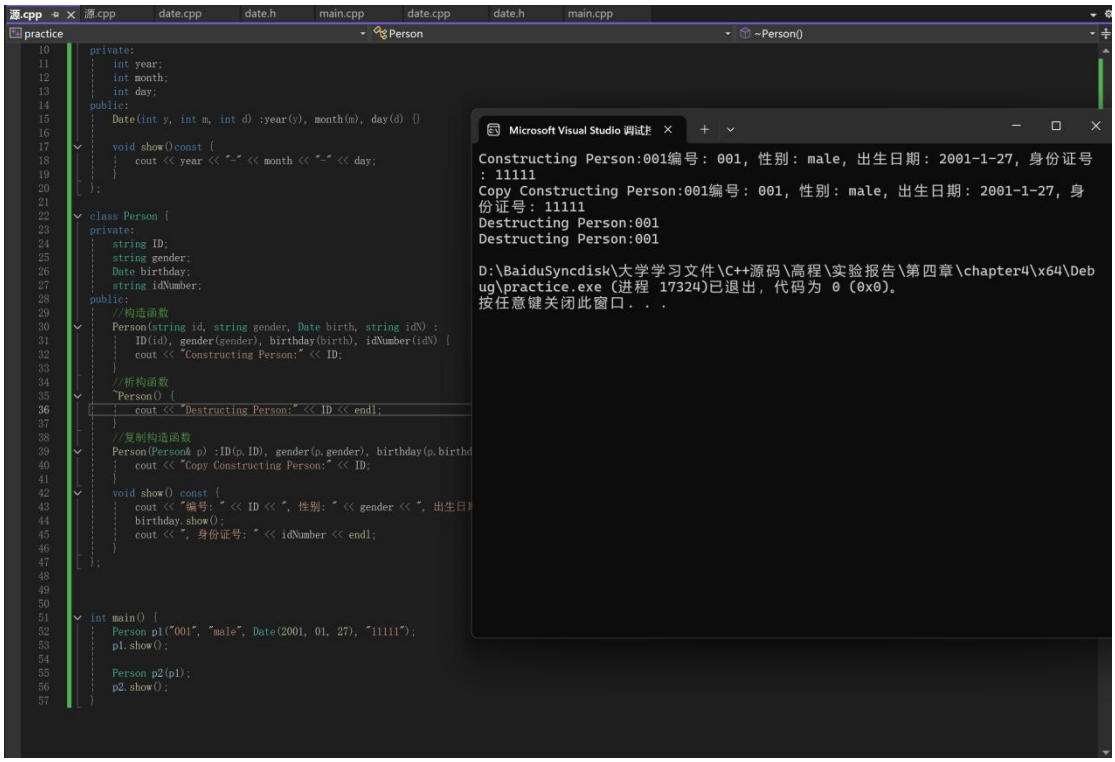
练习 3


// main.cpp
#include "date.h"
int main()
{
Date date1(1999, 4, 20);
cout << "date1:" << endl;
date1.ShowDate();
Date date2(2004, 10, 15);
cout << "date2:" << endl;
date2.ShowDate();
return 0;
}
constructing…
date1:
Date: 1999.4.20
constructing…
date2:
Date: 2004.10.15
destructing…
destructing…
D:\BaiduSyncdisk\大学学习文件\C++源码\高程\实验报告\第四章\chapter4\x64\Debug\Project3.exe (进程 3272)已退出，代码为 0 (0x0)。
按任意键关闭此窗口. . .

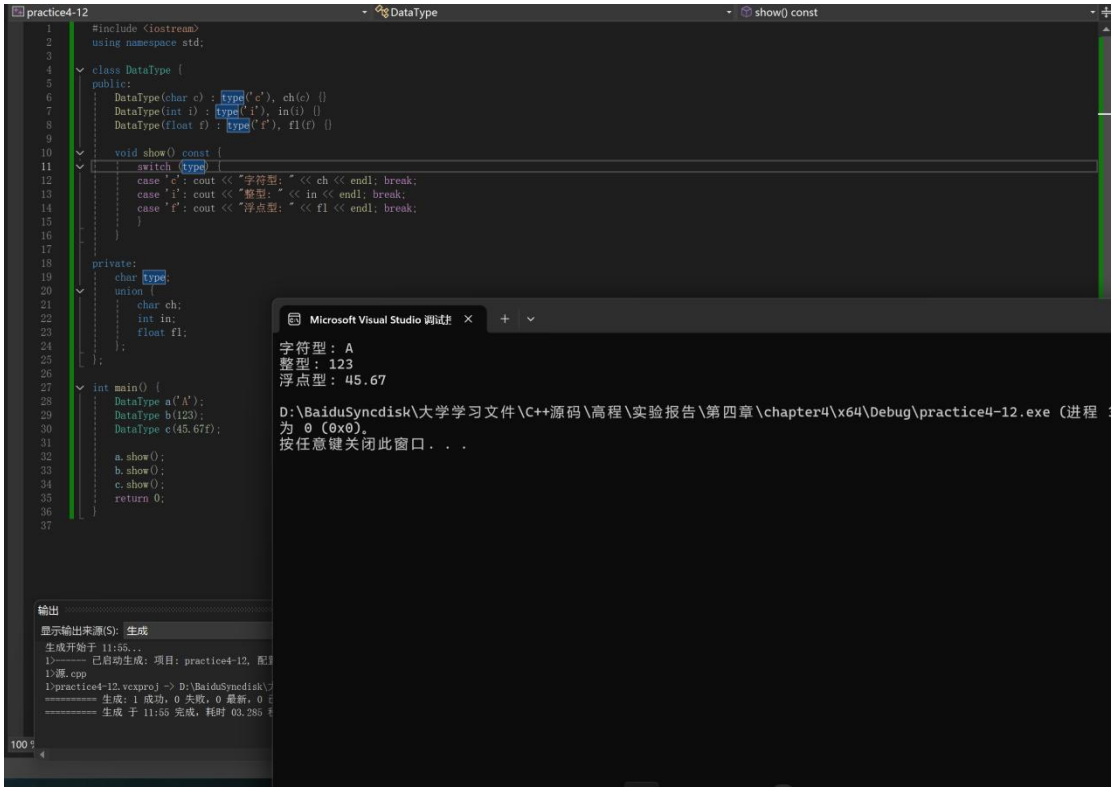
练习 4


class Complex // 定义复数类
{
public:
Complex(double r = 0.0, double i = 0.0); // 带默认参数构造函数
~Complex(); // 析构函数
double abscomplex(); // 求复数的模
private:
double real;
double imag;
};
Complex::Complex(double r, double i)
{
cout << "constructing..." << endl;
real = r;
imag = i;
cout << "real:" << real << ", imag:" << imag << endl;
}
Complex::~Complex()
{
cout << "destructing... real:" << real << ", imag:" << imag << endl;
}
double Complex::abscomplex()
{
double t = real * real + imag * imag;
return sqrt(t);
}
int main()
{
Complex A(1.1, 2.2), B = A;
cout << "abs of complex A=" << A.abscomplex() << endl;
cout << "abs of complex B=" << B.abscomplex() << endl;
return 0;
}
constructing...
real:1.1, imag:2.2
abs of complex A=2.45967
abs of complex B=2.45967
destructing... real:1.1, imag:2.2
destructing... real:1.1, imag:2.2
D:\BaiduSyncdisk\大学学习文件\C++源码\高程\实验报告\第四章\chapter4\x64\Debug\Project4.exe (进程 12028)已退出，代码为 0 (0x0)。
按任意键关闭此窗口. . .

4-10


源.cpp
源.cpp
date.cpp
date.h
main.cpp
date.cpp
date.h
main.cpp
practice
Person
~Person()
private:
int year;
int month;
int day;
public:
Date(int y, int m, int d) :year(y), month(m), day(d) {}
void show()const {
cout << year << "-" << month << "-" << day;
};
class Person {
private:
string ID;
string gender;
Date birthday;
string idNumber;
public:
//构造函数
Person(string id, string gender, Date birth, string idN) :
ID(id), gender(gender), birthday(birth), idNumber(idN) {
cout << "Constructing Person:" << ID;
//析构函数
~Person() {
cout << "Destructing Person:" << ID << endl;
//复制构造函数
Person(Person& p) :ID(p.ID), gender(p.gender), birthday(p.birthd
cout << "Copy Constructing Person:" << ID;
void show() const {
cout << "编号: " << ID << ", 性别: " << gender << ", 出生日
birthday.show();
cout << ", 身份证号: " << idNumber << endl;
};
int main() {
Person p1("001", "male", Date(2001, 01, 27), "11111");
p1.show();
Person p2(p1);
p2.show();
Microsoft Visual Studio 调试
Constructing Person:001编号: 001, 性别: male, 出生日期: 2001-1-27, 身份证号
: 11111
Copy Constructing Person:001编号: 001, 性别: male, 出生日期: 2001-1-27, 身
份证号: 11111
Destructing Person:001
Destructing Person:001
D:\BaiduSyncdisk\大学学习文件\C++源码\高程\实验报告\第四章\chapter4\x64\Deb
ug\practice.exe (进程 17324)已退出, 代码为 0 (0x0)。
按任意键关闭此窗口. . .

4-12


practice4-12
DataType
show() const
#include <iostream>
using namespace std;
class DataType {
public:
DataType(char c) : type('c'), ch(c) {}
DataType(int i) : type('i'), in(i) {}
DataType(float f) : type('f'), fl(f) {}
void show() const {
switch (type) {
case 'c': cout << "字符型: " << ch << endl; break;
case 'i': cout << "整型: " << in << endl; break;
case 'f': cout << "浮点型: " << fl << endl; break;
}
}
private:
char type;
union {
char ch;
int in;
float fl;
};
};
int main() {
DataType a('A');
DataType b(123);
DataType c(45.67f);
a.show();
b.show();
c.show();
return 0;
}
输出
显示输出来源(S): 生成
生成开始于 11:55...
1>------ 已启动生成: 项目: practice4-12, 配
1>源.cpp
1>practice4-12.vcxproj -> D:\BaiduSyncdisk\
========== 生成: 1 成功, 0 失败, 0 最新, 0
========== 生成 于 11:55 完成, 耗时 03.285
Microsoft Visual Studio 调试
字符型: A
整型: 123
浮点型: 45.67
D:\BaiduSyncdisk\大学学习文件\C++源码\高程\实验报告\第四章\chapter4\x64\Debug\practice4-12.exe (进程
为 0 (0x0)。
按任意键关闭此窗口. . .

4-13

```cpp
#include <iostream>
#include <cmath>
using namespace std;

class Circle {
private:
    double radius;
public:
    //构造函数
    Circle(double r) :radius(r) {
        cout << "Constructing Circle:" << radius << endl;
    }
    //析构函数
    ~Circle() {
        cout << "Deconstructing" << endl;
    }
    //计算面积函数
    double getArea()const {
        return 3.14 * radius * radius;
    }
    //输出函数
    void show() const {
        cout << "半径: " << radius << ", 面积: " << getArea() << endl;
    }
};

int main() {
    Circle c(5.0);
    c.show();

    return 0;
}
```

```
Constructing Circle:5
半径: 5, 面积: 78.5
Deconstructing

D:\BaiduSyncdisk\大学学习文件\C++源码\高程\实验报告\第四章\chapter4\x64\Debug\Project4-13.exe (进程 4692)已退出，代码为 0 (0x0)。
按任意键关闭此窗口. . .
```

4-14

```cpp
#include <iostream>
using namespace std;

class CPU {
public:
    CPU(unsigned int freq, unsigned int bits, unsigned int cores, bool ht)
        : frequency(freq), wordLength(bits), coreCount(cores), hyperThreading(ht) {
    }

    void show() const {
        cout << "CPU信息: " << endl;
        cout << "  主频: " << frequency << " MHz" << endl;
        cout << "  字长: " << wordLength << " 位" << endl;
        cout << "  核心数: " << coreCount << endl;
        cout << "  是否支持超线程: " << (hyperThreading ? "是" : "否") << endl;
    }

private:
    //表示所占的位域，也就是二进制位
    unsigned int frequency : 12;     // 支持到 4095 MHz
    unsigned int wordLength : 6;     // 32或64
    unsigned int coreCount : 4;      // 单核/双核/四核
    bool hyperThreading : 1;         // 是否超线程
};

int main() {
    CPU cpu1(2800, 64, 4, true);
    cpu1.show();
    cout << "sizeof(CPU) = " << sizeof(CPU) << " 字节" << endl;
    return 0;
}
```

```
CPU信息:
  主频: 2800 MHz
  字长: 0 位
  核心数: 4
  是否支持超线程: 是
sizeof(CPU) = 8 字节

D:\BaiduSyncdisk\大学学习文件\C++源码\高程\实验报告\第四章\chapter4\x64\Debug\practice4-19.exe (进程 33196)已退出，代码为 0 (0x0)。
按任意键关闭此窗口. . .
```

4-19

```cpp
#include <iostream>
using namespace std;

class CPU {
public:
    CPU(unsigned int freq, unsigned int bits, unsigned int cores, bool ht)
        : frequency(freq), wordLength(bits), coreCount(cores), hyperThreading(ht) {
    }

    void show() const {
        cout << "CPU信息: " << endl;
        cout << "  主频: " << frequency << " MHz" << endl;
        cout << "  字长: " << wordLength << " 位" << endl;
        cout << "  核心数: " << coreCount << endl;
        cout << "  是否支持超线程: " << (hyperThreading ? "是" : "否") << endl;
    }

private:
    //表示所占的位域，也就是二进制位
    unsigned int frequency : 12;     // 支持到 4095 MHz
    unsigned int wordLength : 6;     // 32或64
    unsigned int coreCount : 4;      // 单核/双核/四核
    bool hyperThreading : 1;         // 是否超线程
};

int main() {
    CPU cpu1(2800, 64, 4, true);
    cpu1.show();
    cout << "sizeof(CPU) = " << sizeof(CPU) << " 字节" << endl;
    return 0;
}
```

```
CPU信息:
  主频: 2800 MHz
  字长: 0 位
  核心数: 4
  是否支持超线程: 是
sizeof(CPU) = 8 字节

D:\BaiduSyncdisk\大学学习文件\C++源码\高程\实验报告\第四章\chapter4\x64\Debug\practice4-19.exe (进程 33196)已退出，代码为 0 (0x0)。
按任意键关闭此窗口. . .
```

4-20

```cpp
#include <iostream>
using namespace std;

class Complex {
public:
    Complex(double r = 0.0, double i = 0.0) : real(r), imag(i) {}
    void add(const Complex& c) {
        real += c.real;
        imag += c.imag;
    }
    void show() const {
        cout << real << "+" << imag << "i" << endl;
    }

private:
    double real;
    double imag;//i前面的系数
};

int main() {
    Complex c1(3, 5);    // 用复数初始化
    Complex c2 = 4.5;    // 用实数初始化（隐式转换）
    c1.add(c2);          // 相加
    c1.show();
    return 0;
}
```

```
7.5+5i

D:\BaiduSyncdisk\大学学习文件\C++源码\高程\实验报告\第四章\chapter4\x64\Debug\practice4-20.exe (进程 18564)已退出，代码为 0 (0x0)。
按任意键关闭此窗口. . .
```